Dynamic Emergency Response Allocation Agent

A hybrid agent that combines lookahead planning with conditional LLM arbitration for zero-error emergency dispatch

CS 498 AI Agents in the Wild - Final Presentation

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The Problem: LLMs Can't Optimize Under Hard Constraints

LLMs Excel At

Semantic reasoning, parsing unstructured incident reports, and generating natural language justifications.

LLMs Fail At

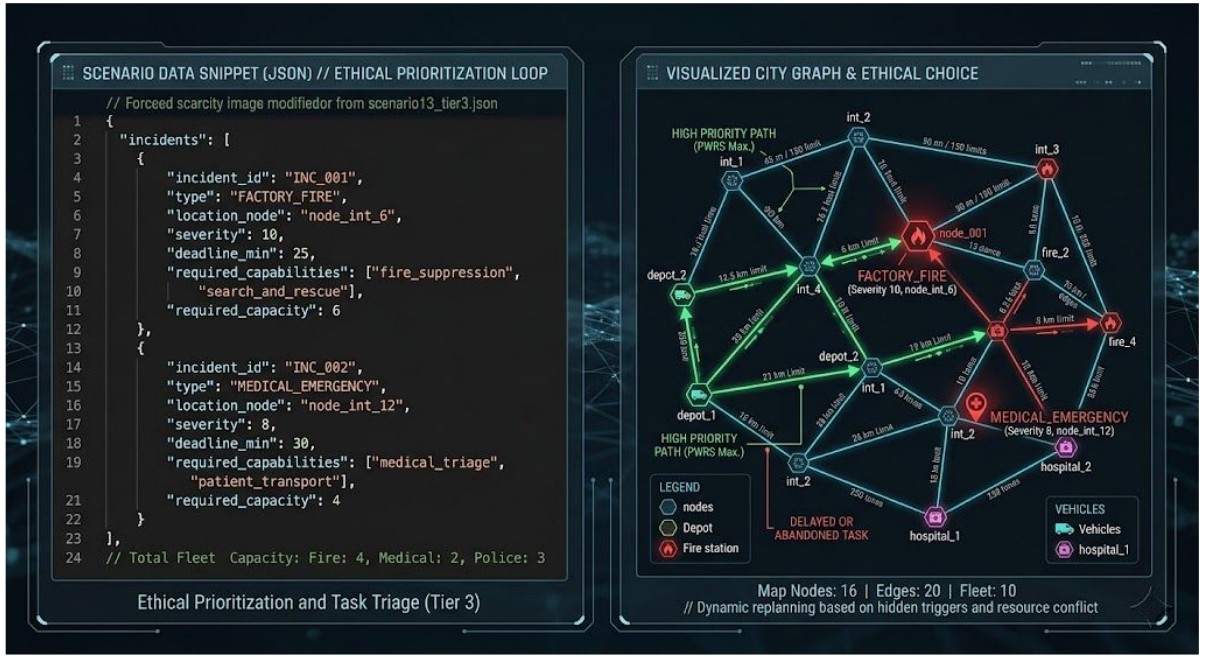
Strict mathematical optimization: they hallucinate routes, lose track of capacities, and output physically infeasible actions.

Disaster Response Requires Both

Zero-error spatial constraint satisfaction *and* unstructured text interpretation - simultaneously, under deadline.

End-to-end LLM solvers fail catastrophically on hard constraints. Our approach (AgentKit): **LLM as orchestrator, not calculator.**

RescueBench: A Deterministic Emergency Dispatch Benchmark



What We Built

- 20 deterministic JSON scenarios on a shared 16-node city map
- Heterogeneous 4-class fleet: ambulances, fire engines, police, supply trucks
- Deadlines verified via Dijkstra's algorithm: no subjective grading

How We Generated It

- Manually authored base city graph guaranteeing solvability
- LLM-assisted mutation pipeline (Gemini) for diverse incidents
- Injected timestamped infrastructure failures: bridge collapses, road floods

Four Tiers, Four Failure Modes

Tier	Challenge	What It Tests
1 - Basic Triage	Direct routing, single-unit dispatches	Sanity check
2 - Constraint Math	Capacity arithmetic, multi-unit coordination	Feasibility reasoning
3 - Ethical Prioritization	Forced scarcity — abandon low-severity to save high-severity	Global optimization
4 - Dynamic Replanning	Mid-mission infrastructure failures triggered by simulation clock	Adaptive replanning

Tier 3 example: Train derailment needs all 4 medical units. Two minor injuries also active. Agent must *intentionally* abandon them to maximize global score.

Tier 4 example: At t=8 min, suspension bridge blocked — agent must recalculate all routes on the fly.

Evaluation & Key Design Decisions

- **Evaluation Metrics:**
  - *Priority-Weighted Resolution Score (PWRS):* Did incidents get solved before the deadline, weighted by severity (1-10)?

    $$S = \sum_{i=1}^{n} (w_i \times I_i)$$

  - *Response Efficiency:* Total agent travel time vs. Dijkstra's absolute mathematical minimum.

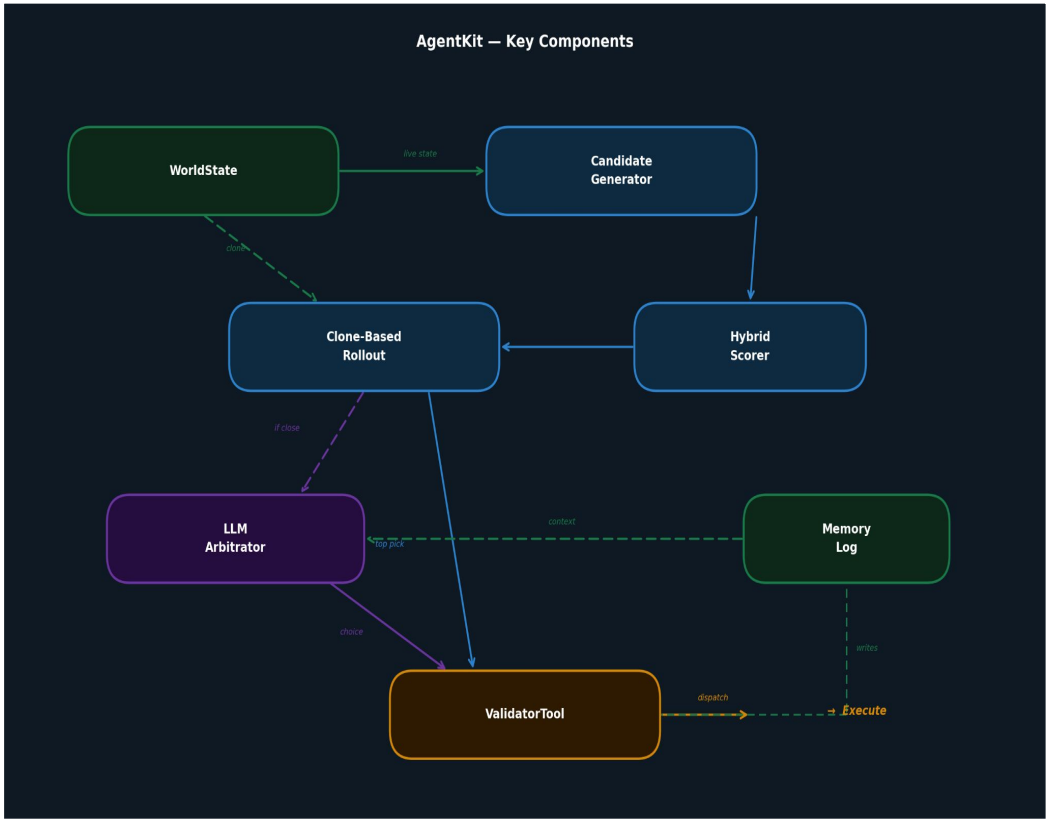
    $$E = \frac{T_{optimal}}{T_{agent}}$$

  - *Constraint Violation Rate:* Hard count of hallucinated, physically impossible actions (e.g., crossing flooded roads).

    $$V = \sum c_{invalid}$$

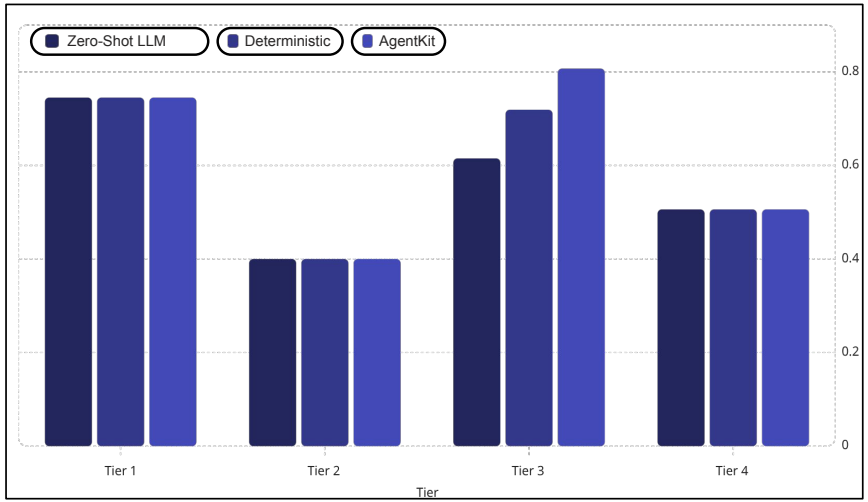
- **Key Design Decisions:**
  - *Deterministic Verifiability:* Completely eliminates subjective "LLM-as-a-judge" grading. Success is governed by Boolean and algebraic Operations Research math.
  - *Forces Heuristic Orchestration:* Proves that LLMs cannot act as raw mathematical calculators; they must use a robust ReAct loop to query external validation tools.

Seven Components, One Hybrid Agent



1	**WorldState** Live simulation — vehicles, incidents, routes, hospitals
2	**Candidate Generator** Enumerates every valid (vehicle → incident) pair
3	**Hybrid Scorer** Ranks candidates by urgency, slack, and scarcity
4	**Clone-Based Rollout** Deep-copies simulator, projects future PWRS per candidate
5	**ValidatorTool** Hard gate — nothing dispatches without passing constraints
6	**LLM (Claude Sonnet 4.5) Arbitrator** Tiebreaker invoked only when top scores are within 8%
7	**Memory Log** Feeds recent decisions and alerts into LLM context window

Results: AgentKit Excels in Tier 3's Prioritization Tradeoffs



Tier	Zero-Shot LLM	Deterministic	AgentKit
1	0.745	0.745	0.745
2	0.400	0.400	0.400
3	0.615	0.719	**0.807**
4	0.506	0.506	0.506
Mean	0.566	0.592	**0.615**

Priority-Weighted Resolution Score (PWRS)

AgentKit achieves the highest mean PWRS (0.615) and is the **only method that never regresses** across any tier.

Tier 3: +8.8pp over deterministic, +19.2pp over zero-shot

- Zero-shot introduces constraint violations on Tier 2
- Tier 4: all methods fail equally — an environmental ceiling

Where AgentKit Wins: Sequencing and Coordination

Tier 3, Scenario 13 — Vehicle Sequencing

One police unit, two incidents: a riot (18.3 min away) and a traffic call (6.7 min away).

Method	Order	PWRS
Zero-Shot	Riot first	0.850
Deterministic	Riot first	0.850
AgentKit	**Traffic → Riot**	**1.000**

Rollout projected that the fast job first returns the unit in time for the riot deadline.

Tier 3, Scenario 15 — Trolley Problem

Three trolley incidents, one police unit, one medic. Zero-shot cannot coordinate parallel dispatch without planning structure.

Method	PWRS
Zero-Shot	0.375
Deterministic	1.000
AgentKit	**1.000**

Where Everyone Struggles: The Infrastructure Ceiling

Tier 4, Scenario 16 — Bridge Collapse Mid-Mission

MED_01 dispatched at t=0. Bridge collapses at t=8, forcing a major detour. Arrival at t=45.3 — far past deadline. **All methods: PWRS = 0.000.** Replanning only affects *future* dispatches: committed vehicles cannot be recalled.

Tier 4, Scenario 19 — Simultaneous Double Blockage

POLICE_01 en route to riot. At t=10: northern approach iced, southern barricaded. After completing the mission, no valid path remains to any open incident. **All methods: PWRS = 0.000.**

Lookahead only helps when *sequencing* matters — Tier 3 gains, not Tier 4. The hard ceiling is infrastructure, not algorithm.

Conclusion: Structure Beats Raw LLM

1. **Pre-validated candidates eliminate violations**

   Zero-shot fails structurally on hard constraints: deterministic grounding is non-negotiable.

2. **Look-Ahead Finds What Greedy Misses**

   Tier 3 PWRS: AgentKit 0.807 vs Rule-Based 0.719 (+8.8 pts from dispatch ordering alone)

3. **Tier 4 is an infrastructure ceiling, not a planning problem**

   When the environment is adversarially constrained, no planning overhead helps.

Future work: Bundle-level joint planning, predictive replanning before dynamic events fire, and stress-testing LLM value in genuinely ambiguous, information-sparse scenarios.